

Skeptical verification of seven deployment claims for dissertation use

Three of seven claims fail independent verification outright; none are fully verified by non-vendor sources. This audit applied a five-tier evidence hierarchy—peer-reviewed studies at the top, vendor press releases at the bottom—and required at least one non-vendor source to mark any claim as verified. The results demand significant rewording throughout the dissertation: two claims (Traffic-R1, Southampton/Minima) should be dropped or relegated to footnotes, one (Iroha 2) contains a factual error that must be corrected, and the remaining four require hedged language that distinguishes vendor assertions from independently confirmed facts.

§1. Methodology: what counts as verification

Evidence was ranked in strict descending order of weight:

1. **Peer-reviewed independent evaluations** — published in journals or conferences, authored by researchers with no vendor affiliation
2. **Operator-published data** — reports from city transport authorities, regulators, or central banks describing their own operational experience
3. **Credible journalism** — FT, WSJ, IEEE Spectrum, MIT Technology Review, Wired, Reuters, Bloomberg, Scientific American, South China Morning Post
4. **Academic reproductions on public datasets** — preprints or benchmarks by independent researchers using publicly available code or data
5. **Vendor press releases and marketing** — flagged as WEAKEST; never accepted alone

A claim earns VERIFIED only if at least one Tier 1–3 source independently corroborates the core assertion. PARTIALLY VERIFIED means non-vendor evidence exists but materially qualifies the claim. VENDOR-ONLY means the only evidence traces back to the claimant. DISPUTED means non-vendor evidence directly contradicts or undermines the claim.

§2. Per-claim verification

Claim 1 — Traffic-R1: an anonymous city and zero independent witnesses

Claim as stated: Traffic-R1 (arXiv:2508.02344, Zou et al., August 2025) is "managing signals affecting over 55,000 drivers daily" in a Chinese city, with queue-length reductions exceeding 5%.

Verification status: UNVERIFIED

The paper explicitly withholds the city name ("name withheld for privacy"), making external corroboration structurally impossible. (arXiv) No Chinese city transport authority, no credible journalist, and no independent academic has referenced this deployment. The only sources repeating the "55,000 drivers" figure are the arXiv paper itself, its HuggingFace model card, and automated paper aggregators—all traceable to the authors. (ResearchGate) (arXiv)

Three of the paper's seven authors are employees of PCITECH (佳都科技, SSE: 600728), a major Chinese intelligent-transportation vendor with revenue exceeding ¥71 billion. The first author holds a dual HKUST(GZ)-PCITECH affiliation. (ResearchGate) This is a vendor-academic collaboration with clear commercial interest, not arm's-length research. The publicly released model checkpoint on HuggingFace is acknowledged as "an earlier checkpoint" that excludes data "to address commercial and privacy concerns," meaning the public artefact differs from the deployed system. (Hugging Face)

The paper's CityFlow simulation benchmarks are theoretically reproducible—code aligns with the LLMLight evaluation framework—(Hugging Face) but **no independent group has published a reproduction** as of April 2026. No evidence exists that the deployment continued past the paper's six-week A/B test window.

Recommended dissertation framing: *"Zou et al. (2025) report a six-week A/B test at ten intersections in an unnamed Chinese city, claiming >5% queue-length reductions for 55,000 daily drivers. The deployment city is anonymised, and no independent verification exists. Three of seven authors are affiliated with PCITECH (SSE: 600728), the ITS vendor that likely operates the platform. The simulation results on public CityFlow datasets have not been independently reproduced."*

Claim 2 — Google Project Green Light: real gains, inflated framing

Claim as stated: 70+ intersections across 17 cities; 30% reduction in vehicle stops; 10% emissions reduction.

Verification status: PARTIALLY VERIFIED

The intersection and city counts are factual but outdated: as of early 2026, Google's own research page lists **20 cities with 47 million monthly car rides**; (Google Research) Boston alone accounts for 114 intersections. (Boston) The headline performance figures, however, are materially weaker than they appear.

The 10% emissions reduction is modelled, not measured. Google's own footnote states it uses "a Department of Energy emissions model" (Google Research) with "a single fuel-based vehicle type... not yet adjusted for local vehicle mix." (Google) The 30% stop reduction is described as

based on "early data points" from Google's own analysis of Google Maps driving patterns—a vendor measuring its own output.

The strongest non-vendor evidence comes from Boston, where the city commissioned INRIX (a third-party traffic analytics firm) to evaluate Phase 2. INRIX found an average **13.5% reduction in delay** and **20% average reduction in unnecessary stops**—materially lower than Google's headline "up to 30%." Transport for Greater Manchester reported **9% improvement in morning peak and 18% in afternoon peak** where recommendations were implemented, (Marketing Stockport) but critically noted that "traffic engineers often opted to ignore Google's recommendations" because signals had been intentionally set to prioritise buses or discourage rat-running through residential areas. (Scientific American)

Seattle DOT reversed at least one Green Light recommendation. SDOT spokesperson Mariam Ali confirmed to Scientific American (August 2024) that a timing change was reverted because it "did not result in a net benefit"—(Scientific American) it improved one direction of travel but degraded overall intersection performance. The specific intersection was not named.

Two academic traffic engineers have critiqued the methodology on the record. **Henry Liu** (University of Michigan) told Scientific American: "It's all dependent on the baseline you're comparing to... most [pilot intersections] aren't currently running adaptive systems." (Scientific American) **Aleksandar Stevanovic** (University of Pittsburgh) warned the approach mirrors highway-expansion logic and risks induced demand. (Scientific American) Google project manager Matheus Vervloet acknowledged that test intersections were "selected specifically to avoid complicating factors" like bus lanes, bike lanes, and pedestrian crossings—** (Google) a confirmed selection effect**. (Scientific American)

No peer-reviewed independent evaluation of Project Green Light exists. No participating city outside Boston has published its own quantitative assessment. For Hamburg, Rio de Janeiro, Jakarta, Bengaluru, Abu Dhabi, Haifa, and Budapest, the only available statements are vendor-curated quotes on Google's own pages.

Recommended dissertation framing: "Google reports that Project Green Light has expanded to 20 cities as of 2025, with modelled estimates of up to 30% stop reduction and 10% emissions reduction. An INRIX third-party evaluation in Boston found a 13.5% average delay reduction and 20% stop reduction—below Google's headline figures. (Boston) Academic critics note that pilot intersections were selected to avoid complexity and were compared against outdated fixed-time signals, not modern adaptive systems (Liu, Scientific American 2024). The emissions figure is modelled using a single vehicle type and has not been independently measured. At least one recommendation was reversed by Seattle DOT for failing to deliver a net benefit."

Claim 3 — Alibaba City Brain: a ranking measured by Alibaba's own subsidiary

Claim as stated: Hangzhou dropped from 5th most congested city to 57th in China's rankings

thanks to City Brain.

Verification status: DISPUTED

The congestion ranking comes from AutoNavi (Gaode Maps), which Alibaba acquired [\(Wikipedia\)](#) **for \$1.5 billion in 2014.** [\(South China Morning Post\)](#) The "5th to 57th" claim was announced by Alibaba Cloud President Simon Hu at Alibaba's own Computing Conference in September 2018 and disseminated via Alizila, Alibaba's news portal. Every academic paper citing this statistic traces back through the same circular chain to Alibaba's 2018 press conference. No independent ranking source (TomTom, INRIX, Baidu Maps) corroborates the specific claim.

The baseline "5th" figure is itself imprecise. AutoNavi's own published annual reports show Hangzhou ranked **#2 in 2014–2015 and #8 in 2016**—never exactly 5th in any annual dataset located. Additionally, AutoNavi expanded its city coverage from 38 to over 100 cities during this period, mechanically shifting every city's ranking.

Attribution to City Brain is confounded by at least four concurrent interventions. Hangzhou implemented a strict licence-plate lottery and auction system in March 2014 [\(China Daily\)](#) (two years before City Brain launched), capping new registrations at ~80,000 per year. [\(OilPrice.com\)](#) Peak-hour driving restrictions based on plate numbers had been in effect since 2011.

[\(Shanghai Daily\)](#) Approximately **90 km of new metro track** opened between 2012 and 2017 (Lines 1, 2, and 4). [\(Grokikipedia\)](#) Alibaba's own Dr Wang Jian acknowledged at the 2018 conference that "20% of the roads in Hangzhou are under construction." [\(Alizila\)](#) No study has attempted to isolate City Brain's specific contribution from these simultaneous changes.

City Brain remains operational—version 3.0 launched in March 2025, integrating the DeepSeek-R1 model—[\(Hangzhou Government\)](#) but no independent evaluation of its traffic impact has been published. The system is now far broader than traffic optimisation, encompassing [\(Wiley Online Library\)](#) AI policing, healthcare, and geological monitoring. [\(Hangzhou Government\)](#) [\(China Daily\)](#)

Recommended dissertation framing: *"Alibaba reported at its 2018 Computing Conference that Hangzhou's congestion ranking improved from approximately 5th to 57th on AutoNavi's index. However, AutoNavi is a wholly owned Alibaba subsidiary, [\(Wikipedia\)](#) [\(BridgingNews\)](#) creating a material conflict of interest. The improvement period (c. 2016–2018) coincided with licence-plate restrictions (from 2014), [\(China Daily\)](#) new metro lines (~90 km added 2012–2017), [\(Grokikipedia\)](#) and peak-hour driving bans (from 2011). No independent study has isolated City Brain's contribution from these concurrent interventions, and no non-Alibaba source has verified the specific ranking figures."*

Claim 4 — Southampton/Minima blockchain drone: extraordinary claims, no evidence

Claim as stated: World's first blockchain black box on autonomous drone hardware during live

flight; 500x performance gain; 10,000% energy efficiency gain.

Verification status: VENDOR-ONLY — recommend dropping from dissertation

This claim rests entirely on a University of Southampton press release (March 2026) and promotional material from Minima, a blockchain startup with its own cryptocurrency token (MINIMA, ~\$0.03 market cap ~\$12M). **No peer-reviewed paper, no reproducible code, no specified performance baseline, and no independent technical review exist.**

The "500x performance gain" and "10,000% energy efficiency gain" lack any defined baseline. The press release states these gains come from "moving the technology from software into hardware" but never specifies what software, on what hardware, measuring what metric.

(University of Southampton) (The Engineer) In FPGA/ASIC literature, large speedups over unoptimised software are routine and uninformative without a rigorous baseline. The exact ARM chip model used is never named. No flight parameters (duration, altitude, data volume, number of blockchain nodes) are disclosed.

This was an undergraduate student group design project completed in a single semester,

(University of Southampton) supervised by Dr Ivan Ling, whose publication record is in sensors and nanocrystalline graphene—not blockchain or distributed systems. (University of Southampton)

(Interesting Engineering) The October 2025 partnership announcement promised "independent assessment from regulators, UAV manufacturers, and IoT security experts" in Q1 2026; (DL News) (Chainwire) no such assessment has appeared. Prior academic work (Böhm, Distler & Kapitza, ROBUST '25) has explored blockchain-based drone flight recording, (Gi-robust) questioning the "world's first" characterisation.

Zero credible tech publications (IEEE Spectrum, The Register, Ars Technica, Wired) covered this. All media coverage consists of press-release rewrites distributed via Chainwire, a paid crypto PR wire service.

Recommended dissertation framing: Drop this claim entirely. If it must be mentioned: *"The University of Southampton issued a press release in March 2026 describing an undergraduate student project that demonstrated a Minima blockchain node on drone hardware, claiming 500x performance and 10,000% energy efficiency gains over an unspecified software baseline. No peer-reviewed publication, reproducible methodology, or independent assessment exists. Minima is a blockchain startup (Deep Analysis) with a cryptocurrency token, (Crunchbase) creating commercial incentive for promotional claims."*

Claim 5 — Yunex Traffic FUSION at TfL: real trials, aspirational numbers

Claim as stated: Up to 14% delay reduction; deployment target 4,350 sites by 2030/31.

Verification status: PARTIALLY VERIFIED

The 4,350-site target and timeline are confirmed by TfL's own "London on the Move" strategy document (January 2026), which states a target of 500 sites by end of FY2025/26 and full rollout to 4,350 by 2030/31. (tfl) **Current deployment is therefore in the low hundreds at most, not thousands.**

The 14% figure is real-world data, but from the best-performing site during peak hours only. A Yunex Traffic white paper reports results from a Hampshire County Council trial (February–May 2023) where FUSION was tested against calibrated SCOOT at three locations. St Margaret's Roundabout achieved **14% journey-time improvement during the PM peak**, but only **8% all-day**. (yunextraffic) Ash Road Corridor showed 5% all-day. Havant Town Centre showed 4–7%. (yunextraffic) **The all-day average across sites was 4–8%, not 14%.** TfL's own language is carefully hedged: "The changes *could* reduce delays by up to 14 per cent." (CiTTi Magazine)

The dissertation should correct the corporate attribution. Siemens sold Yunex Traffic to Atlantia (now Mundys S.p.A., Benetton/Blackstone ownership) for €950 million in June 2022. (Mundys) (Siemens) Yunex Traffic has no remaining Siemens affiliation. (Yunex Traffic) The original SCOOT-replacement timeline has slipped substantially—a 2020 plan targeted full deployment by 2023.

No independent peer-reviewed evaluation of FUSION exists. The Hampshire trial paper was authored by Yunex Traffic staff. An ITS World Congress 2022 paper reporting 15% improvement over SCOOT at a London "Living Lab" of ~15 junctions was also vendor-authored. (yunextraffic) No academic or consultant audit was found.

Recommended dissertation framing: *"Yunex Traffic's FUSION adaptive signal control, deployed under contract to TfL, achieved 4–14% journey-time improvements in a real-world Hampshire trial (2023), with the 14% figure representing peak-period performance at the best-performing site. TfL targets 500 FUSION sites by end of 2025/26 and 4,350 by 2030/31, replacing SCOOT. All published evaluations are vendor-authored; no independent audit exists. Note: Yunex Traffic was sold by Siemens to Mundys S.p.A. in 2022 and is no longer a Siemens entity."*

Claim 6 — Hyperledger Iroha 2: a version-number error that undermines the argument

Claim as stated: Iroha 2 is "IoT-ready," designed for resource-constrained environments, with ~1s finality. Named production users include Cambodia's Bakong, Moscow Exchange/D3 Ledger.

Verification status: DISPUTED — contains factual errors

Cambodia's Bakong runs Iroha 1, not Iroha 2. This is confirmed by Iroha 2 Technical Lead Aleksandr Petrosyan (Soramitsu): "Bakong is running Iroha v1 to this day." (Medium) The Hyperledger Foundation's own February 2025 announcement carefully distinguishes Bakong (Iroha 1, production since 2020) from Iroha 2 projects, which are all proof-of-concept: Bokolo Cash (Solomon Islands), Digital Kina (Papua New Guinea), and Palau Invest. (Ifdecentralizedtrust)

Bakong's impressive scale—30M+ wallets, 608M+ transactions, \$150B+ processed in 2024—belongs to Iroha 1. (Soramitsu)

Moscow Exchange/D3 Ledger also runs Iroha 1, identifiable by its use of YAC consensus (Iroha 1's algorithm; Iroha 2 uses Sumeragi). D3 Ledger's pilot comprised only 3 nodes. (Forklog) Furthermore, Russia's National Settlement Depository was sanctioned by the EU (June 2022) and US OFAC (June 2024), making this deployment effectively defunct for international purposes. (Harneys)

Iroha 2 is not production-stable. The latest release is v2.0.0-rc.2.1 (release candidate). (GitHub) The February 2025 announcement called it an "MVP release." (Ifdecentralizedtrust) Quarterly reports to the Hyperledger Technical Oversight Committee note "challenges related to a shortage of high-quality and available development resources," and the maintenance team "currently comprises SORAMITSU employees" with minimal external contributors. (Hyperledger TOC)

The ~1s finality claim is not supported even by vendor documentation. Soramitsu founder Makoto Takemiya stated a goal of "about two seconds." (Altos) The Hyperledger Architecture White Paper describes Sumeragi as providing "finality within seconds" (plural).

(Hubspotusercontent-na1) Bakong documentation states "less than 5 seconds." (The Asian Banker) No peer-reviewed benchmark of Iroha 2's consensus performance exists. No independent benchmark on constrained hardware (Raspberry Pi, Jetson) was found for either version.

Recommended dissertation framing: *"Hyperledger Iroha 2 remains at release-candidate stage (v2.0.0-rc.2.1) with no confirmed large-scale production deployment. All major production systems attributed to 'Iroha'—including Cambodia's Bakong CBDC (30M+ wallets) and D3 Ledger—run Iroha 1, not Iroha 2, as confirmed by the Iroha 2 technical lead. (Medium) Vendor documentation describes finality 'within seconds,' not the ~1s sometimes cited. No independent benchmark of Iroha 2 on resource-constrained hardware has been published."*

Claim 7 — NVIDIA Jetson Orin Nano Super: broadly correct, engine-dependent

Claim as stated: 20–40 tok/s for 3B INT4 models per JetPack 6.2 benchmarks.

Verification status: PARTIALLY VERIFIED

NVIDIA's official benchmarks (December 2024, January 2025) show **Llama 3.2 3B at 43.07 tok/s** and **Phi-3.5 3.8B at 38.1 tok/s** using MLC (Machine Learning Compilation) API with INT4 quantization (q4f16_ft format) (nvidia) in MAXN 25W power mode with (jetson_clocks) enabled. The claimed "20–40" range actually **understates** 3B model performance while reflecting 7–8B models (Llama 3.1 8B: 19.14 tok/s). (nvidia)

Independent community benchmarks consistently show 7–15% lower numbers. Developer forum user mohamed.alsalti achieved 39.94 tok/s on Llama 3.2 3B after correct configuration—

[nvidia](#) 7% below NVIDIA's figure. User gang.cheng measured 19.23 tok/s on Qwen2.5 7B versus NVIDIA's 21.75—12% below. [nvidia](#) These gaps are consistent and suggest NVIDIA's published figures represent optimistic but not unrealistic conditions.

The inference engine is the critical variable. NVIDIA's benchmarks use MLC, a highly optimised compiler. Users running Ollama (llama.cpp backend) or vLLM will see substantially different results—Eric X. Liu's roofline analysis shows Ollama outperforms vLLM by 2–6x on this hardware, [Ericxliu](#) [ericxliu](#) but neither matches MLC for small models. A Medium developer report documents a failure mode where a 3B model silently fell back to CPU inference at **0.3 tok/s** due to unified-memory pressure. [Medium](#) Thermal throttling under sustained load is acknowledged by NVIDIA's own documentation, which calls MAXN mode "experimental." [NVIDIA](#)

Recommended dissertation framing: "NVIDIA's official benchmarks report 38–43 tok/s for 3B-class INT4 models on the Jetson Orin Nano Super using MLC inference engine at 25W. Independent community reproductions confirm the general range but consistently measure 7–15% lower throughput. Performance is highly sensitive to inference engine choice (MLC vs llama.cpp vs vLLM), power mode, and thermal conditions. The '20 tok/s' lower bound applies to 7–8B models, not 3B."

§3. Summary table

#	Claim	Status	Best non-vendor evidence	Key weakness
1	Traffic-R1: 55,000 daily drivers, >5% queue reduction	UNVERIFIED	None found	Anonymous city; 3/7 authors from vendor (PCITECH SSE:600728); zero independent coverage
2	Google Green Light: 30% stop reduction, 10% emissions cut	PARTIALLY VERIFIED	INRIX evaluation (Boston): 13.5% delay, 20% stops; Boston Scientific American interviews with academic critics	Emissions modelled not measured; Google selection bias acknowledged; Manchester engineers often ignored recommendations Scientific American

#	Claim	Status	Best non-vendor evidence	Key weakness
3	Alibaba City Brain: Hangzhou 5th → 57th congestion	DISPUTED	None independent; ranking source is Alibaba subsidiary AutoNavi South China Morning Post BridgingNews	Confounded by metro expansion, plate restrictions, driving bans; circular citation chain
4	Southampton/Minima drone blockchain: 500x, 10,000% gains	VENDOR-ONLY	None found	Undergrad project; no peer review; no baseline specified; crypto startup partner with token
5	Yunex FUSION at TfL: 14% delay reduction, 4,350 sites	PARTIALLY VERIFIED	TfL "London on the Move" confirms targets; Hampshire real-world trial data exists	14% is peak-period best-site; all-day average 4–8%; all evaluations vendor-authored
6	Iroha 2: IoT-ready, ~1s finality, Bakong deployment	DISPUTED	Iroha 2 tech lead confirms Bakong runs Iroha 1; LF Decentralized Trust blog distinguishes versions	Bakong/D3 Ledger are Iroha 1; Iroha 2 still at release candidate; ~1s finality unsupported
7	Jetson Orin Nano Super: 20–40 tok/s for 3B INT4	PARTIALLY VERIFIED	Community benchmarks at 7–15% below NVIDIA figures; multiple independent reproductions	Specific to MLC engine; thermal throttling risk; 20 tok/s applies to 7B not 3B models

§4. Claims to drop or substantially revise

Drop entirely:

- **Claim 4 (Southampton/Minima drone)** — No peer review, no reproducible methodology, no specified baseline, no independent coverage. An undergraduate semester project [University of Southampton](#) promoted via crypto PR wire is not dissertation-grade evidence. If mentioned at all, relegate to a footnote acknowledging it as an unverified industry announcement.

Drop or relegate to a footnote:

- **Claim 1 (Traffic-R1)** — An anonymised city, vendor-affiliated authors, and zero independent corroboration make this claim impossible to verify by design. It can be mentioned as an example of emerging LLM-based signal control research, but should not be cited as evidence of production deployment.

Correct factual errors immediately:

- **Claim 6 (Iroha 2)** — The dissertation must not state or imply that Bakong, D3 Ledger, or Moscow Exchange run Iroha 2. These are confirmed Iroha 1 deployments. (Medium) Iroha 2 has no large-scale production user. (Hyperledger TOC) The ~1s finality figure should be removed or corrected to "within seconds" per vendor documentation.

Retain with mandatory hedging:

- **Claims 2, 3, 5, and 7** may remain but must use the recommended framing language above. Each requires explicit acknowledgement of vendor-origin data, selection effects, confounding factors, or engine-specific conditions as applicable. The dissertation should never present Google's "30% / 10%" or Alibaba's "5th to 57th" as independently established facts.